

Problem

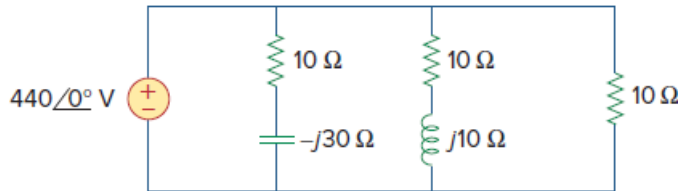
Consider the power system shown in Fig. 11.90. Calculate:

(a) the total complex power

(b) the power factor

(c) the parallel capacitance necessary to establish a unity power factor

Figure 11.90:



Step-by-step solution

Step 1 of 9

Refer to Figure 11.90 in the text book.

Consider the circuit diagram shown in Figure 1.

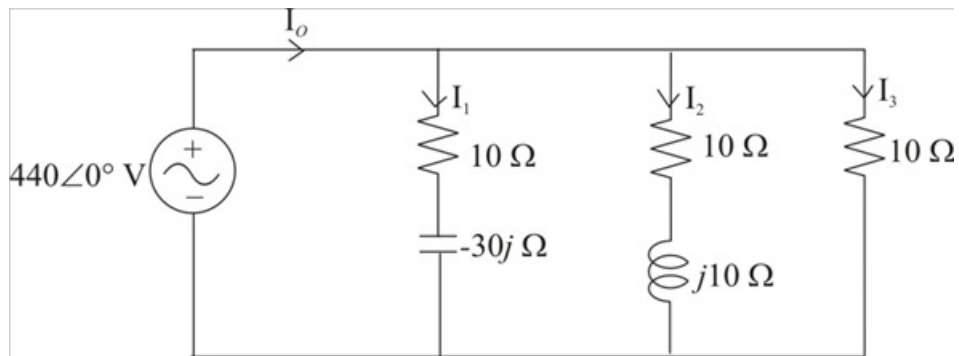


Figure 1

Step 2 of 9

Calculate the value of the current I_1 .

$$\begin{aligned} I_1 &= \frac{440\angle 0^\circ}{10 - j30} \\ &= \frac{440\angle 0^\circ}{31.62\angle -71.57^\circ} \\ &= 13.92\angle 71.57^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_1 is, $13.92\angle 71.57^\circ \text{ A}$.

Step 3 of 9

Calculate the value of the current I_2 .

$$\begin{aligned} I_2 &= \frac{440\angle 0^\circ}{10 + j10} \\ &= \frac{440\angle 0^\circ}{14.14\angle 45^\circ} \\ &= 31.12\angle -45^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_2 is, $31.12\angle -45^\circ \text{ A}$.

Step 4 of 9

Calculate the value of the current I_3 .

$$\begin{aligned} I_3 &= \frac{440\angle 0^\circ}{10} \\ &= 44\angle 0^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_3 is, $44\angle 0^\circ \text{ A}$.

Step 5 of 9

Calculate the value of the current I_o .

$$I_o = I_1 + I_2 + I_3$$

Substitute $44\angle 0^\circ \text{ A}$ for I_3 , $31.12\angle -45^\circ \text{ A}$ for I_2 and $13.92\angle 71.57^\circ \text{ A}$ for I_1 .

$$\begin{aligned} I_o &= 13.92\angle 71.57^\circ + 31.12\angle -45^\circ + 44\angle 0^\circ \\ &= 70.95\angle -7.12^\circ \text{ A} \end{aligned}$$

Therefore, the value of the total current I_o is $70.95\angle -7.12^\circ \text{ A}$.

Step 6 of 9

(a)

Calculate the value of the complex power S .

$$S = VI_o^*$$

Substitute $70.95\angle -7.12^\circ \text{ A}$ for I_o and $440\angle 0^\circ \text{ V}$ for V .

$$\begin{aligned} S &= (440\angle 0^\circ)(70.95\angle -7.12^\circ)^* \\ &= (440\angle 0^\circ)(70.95\angle 7.12^\circ) \\ &= 31.218\angle 7.12^\circ \text{ KVA} \end{aligned}$$

Therefore, the value of the total complex power S is $31.218\angle 7.12^\circ \text{ KVA}$.

And,

$$S = P + jQ$$

$$31.218\angle 7.12^\circ = P + jQ$$

$$30.98 \text{ kW} + j3.869 \text{ kVAR} = P + jQ$$

Hence, the real power is 30.98 kW , and the reactive power is 3.869 kVAR .

Step 7 of 9

(b)

Calculate the value of the power factor.

$$\text{pf} = \cos(\theta_v - \theta_i)$$

Substitute 0° for θ_v and -7.12° for θ_i .

$$\begin{aligned}\text{pf} &= \cos(\theta_v - (-7.12^\circ)) \\ &= \cos 7.12^\circ \\ &= 0.9923\end{aligned}$$

Therefore, the value of the power factor is 0.9923.

Step 8 of 9

(c)

For the circuit to operate at unity power factor, the value of the reactive power should be suppressed to zero by inserting a capacitor across the load.

Step 9 of 9

Calculate the value of the capacitor that produces reactive power of 3.869 kVAR .

$$C = \frac{Q}{2\pi f V^2}$$

Substitute 3.869 kVAR for Q , 60 Hz for f and $440 \text{ V}_{\text{rms}}$ for V .

$$\begin{aligned}C &= \frac{3.869(10^3)}{2\pi(60)(440)^2} \\ &= \frac{3.869(10^3)}{2\pi(60)(193600)} \\ &= 53.01 \mu\text{F}\end{aligned}$$

Therefore, the required value of the capacitor C to operate the circuit at unity power factor is

53.01 μF .